Mahmoud Alyosify

Specialised AI Research Engineer | Efficient LLM Inference & Rigorous Empirical Evaluation | MSc Artificial Intelligence, Queen's University (Canada)

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PROFESSIONAL SUMMARY

AI Research Engineer whose work is defined by **experimental rigour**: pre-registered protocols, honest baselines, ablations and reproducibility on pinned hardware. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**; master's research introduces a **training-free decoding-time method** for token-efficient LLM inference, validated across **three model families** and **four benchmarks** under **paired non-inferiority testing**. Additional empirical work spans a **42-experiment** self-supervised augmentation study, a **GitHub-mining study of human versus agentic code review**, and deep-RL benchmarking against realistic baselines. Comfortable owning a research question end to end — literature, hypothesis, implementation, evaluation, and a result that survives scrutiny.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Research Project (CISC 898):** *Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines**.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — contactless vital-sign estimation via rPPG. **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**; **Finalist, ISEIC 2023**.

RESEARCH & TECHNICAL SKILLS

Research Methodology	Pre-Registered Evaluation Protocols, Paired Non-Inferiority Testing, Statistical Hypothesis Testing, Ablation Studies, Controlled Augmentation Sweeps, Baseline Design, Error Analysis, Deterministic & Reproducible Experimentation, Scientific Writing
Research Areas	Efficient LLM Inference & Decoding, Self-Supervised & Contrastive Learning, Deep Reinforcement Learning, Multi-Agent Systems, Empirical Software Engineering, Applied Predictive Modelling
Benchmarks & Metrics	MT-Bench, YapBench, XSum, SQuAD-v2, BERTScore, ROUGE, Exact-Match / F1, Length-Controlled LLM-as-Judge, Top-1 / Retrieval Metrics, TTFT & Inter-Token Latency, GPU Energy (NVML)
Frameworks	PyTorch, Hugging Face Transformers / PEFT / Datasets, TensorFlow, Scikit-learn, Gymnasium, Optuna, SHAP, NumPy, Pandas, PySpark
Infrastructure	AWS GPU Instances, Docker, ONNX, FAISS, FastAPI, Git/GitHub, Linux, CI/CD
Programming	Python, C++, C, SQL, C#, Go, MATLAB, Bash

RESEARCH PROJECTS

Verbosity-Aware Decoding for Token-Efficient Language Model Inference — *2026 – Present*
MSc Research Project (CISC 898), Queen's University · Supervisor: Dr. Rahatara Ferdousi

- **Contribution:** a **training-free LogitsProcessor** that reduces generated tokens at **decoding time** by fusing stop-token, n-gram-repetition and semantic-redundancy signals under an **entropy-driven gate** — requiring no retraining, no architecture change and adding no KV-cache or attention overhead.
- **Generality:** validated across **three open-weight model families** (Mistral-7B-Instruct, Llama-3.1-8B-Instruct, Qwen2.5-3B-Instruct) and **four benchmarks** (MT-Bench, YapBench, XSum, SQuAD-v2) spanning open-ended chat, summarisation and extractive QA.
- **Rigour:** designed a **pre-registered protocol** that tests semantic preservation as a **paired non-inferiority hypothesis** (BERTScore, ROUGE, exact-match/F1, length-controlled LLM judge) rather than reporting a favourable mean — alongside token reduction, TTFT / inter-token latency and **GPU energy** via NVML.
- **Reproducibility:** fully deterministic runs on pinned AWS GPU instances with fixed seeds and versioned environments.

Analyzing Review Effort: Human vs. Agentic Pull Requests — *2026*
Empirical Software Engineering · Repository Mining · Statistical Analysis

- Built a **reproducible GitHub-mining pipeline** to compare code-review effort on agent-generated versus human-authored pull requests across large-scale open-source repositories.
- Framed the study around a measurable question — does AI-generated code shift reviewer burden? — with defined inclusion criteria, comparable cohorts and quantified outcome measures.

SimCLR-Vision-SSL — Controlled Study of Augmentation in Contrastive Learning — *2026*
PyTorch · SimCLR · SupCon · ONNX · FAISS

- Ran a **42-experiment augmentation sweep** isolating which transformations drive representation quality, then extended the setup with a semi-supervised **SupCon** objective, reaching **84.30% top-1 accuracy** across classification and retrieval evaluation.

Deep RL for Cloud Resource Provisioning — Benchmarked Against Real Baselines *2026*
PPO · DQN · Custom Gymnasium Environment

- Built a custom simulation environment and compared **PPO** and **DQN** on a multi-objective cost–latency reward against **static** and **threshold-based** autoscaling policies actually used in practice.

Further Applied Research

- **HORUS Sentinel** — multi-agent OSINT platform correlating findings into a knowledge graph reasoned over by a self-hosted fine-tuned LLM.
- **SCRAP** — predictive modelling under an operationally constrained **48-hour** lead-time window.
- **Horus-OSINT** — domain adaptation of **Meta-Llama-3-8B** on **159,826** GTD/GDELT records, deployed on AWS.

TEACHING & PROFESSIONAL EXPERIENCE

Machine Learning Instructor *Jul 2025 – Sep 2025*
National Telecommunication Institute (NTI) Cairo, Egypt

- Designed and delivered a **90-hour** graduate-level ML program covering supervised learning, PCA and neural networks with hands-on labs.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Authored Arabic-language curricula in discrete mathematics, algorithms, probability and graph data mining for **10,000+ learners** (**700K+ views**).

AI in Cybersecurity Intern *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) Cairo, Egypt

- Applied AI-driven tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles**; co-developed an internal data-management application in **C#, .NET, SQL Server**.

AWARDS, CERTIFICATIONS & LANGUAGES

Digilians Presidential Scholarship — Queen's University	**Semi-Finalist** — Microsoft Imagine Cup 2023
Finalist — ISEIC 2023	**1st Place** — Smart Cities Hackathon 2022
Ideal Student Award — Assiut University, 2022–2023	**Machine Learning Specialization** — DeepLearning.AI
AWS Academy — ML Foundations, ML for NLP, Data Engineering	**HCIP – AI** (Huawei)

- **Languages:** Arabic (Native) · English (Professional Working Proficiency) · French (Basic).